

# Animal Species Recognition using Deep Learning

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## Abstract

Animal species recognition using image classification has become an important application of deep learning in recent years. It plays a major role in areas such as wildlife monitoring, biodiversity conservation, environmental studies, and automated surveillance systems. Traditional image classification methods usually depend on manual feature extraction, which often struggles to perform well when images have complex backgrounds or varying conditions.

This project presents an animal species recognition system using deep learning techniques on the Animals-10 dataset. Two different models were developed and compared: a Custom Convolutional Neural Network (CNN) and a transfer learning model based on MobileNetV2. Before training, the dataset was preprocessed using image resizing, normalization, and data augmentation to improve model performance and reduce overfitting.

Both models were evaluated using validation accuracy, validation loss, precision, recall, F1-score, and confusion matrix analysis. The experimental results showed a clear difference in performance between the two approaches. The Custom CNN achieved a validation accuracy of 67.64

The results indicate that MobileNetV2 performs significantly better than the Custom CNN in classifying animal species. This demonstrates that transfer learning provides stronger feature extraction and better generalization, making it a more effective solution for multi-class animal image classification tasks.

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# 1 Introduction

Image classification is one of the most widely used applications of deep learning and computer vision. It allows machines to identify and classify objects from images with high accuracy. In recent years, image classification has become highly important in many real-world applications, including medical diagnosis, security systems, agriculture, and wildlife monitoring.

Animal species recognition is a practical application of image classification that focuses on identifying different animal species from images. This task is useful in wildlife conservation, biodiversity monitoring, ecological research, and automated surveillance systems. Accurate animal recognition helps researchers and conservationists collect data efficiently without relying entirely on manual observation.

Traditional machine learning methods often require manual feature extraction, which can be time-consuming and less effective for complex image datasets. Deep learning, especially Convolutional Neural Networks (CNNs), has significantly improved image classification performance by automatically learning important visual features directly from images.

This project focuses on classifying animal images into 10 different species using deep learning techniques. Two models were implemented and compared: a Custom CNN model and MobileNetV2 using transfer learning. The main objective of this study is to compare the performance of both models and determine which approach is more effective for animal species classification.

The performance of both models was evaluated using multiple metrics such as accuracy, loss, precision, recall, and confusion matrix analysis. Based on the experimental results, MobileNetV2 achieved significantly better performance than the Custom CNN model, showing the advantage of transfer learning in image classification tasks.

## 2 Related Work

Deep learning has transformed the field of image classification over the last decade. Among various deep learning techniques, Convolutional Neural Networks (CNNs) have become the most popular approach for extracting visual features and solving classification problems with high accuracy.

Many researchers have applied CNN-based models to animal recognition and wildlife monitoring tasks. These models can automatically learn important image patterns such as shape, texture, and color, making them highly effective for species classification.

Several pre-trained deep learning architectures, including VGGNet, ResNet, EfficientNet, and MobileNet, have achieved strong performance in image recognition benchmarks. These models are widely used in transfer learning because they can reuse knowledge learned from

large datasets and adapt efficiently to new classification tasks.

Transfer learning has become especially useful when working with limited datasets or when training time and computational resources are constrained. Instead of building a model entirely from scratch, pre-trained models allow faster training and often produce better results.

Among transfer learning models, MobileNetV2 is known for its lightweight architecture and efficient performance. It provides a good balance between computational efficiency and classification accuracy, making it suitable for real-world applications.

In this project, MobileNetV2 was selected as the transfer learning model and compared with a Custom CNN architecture. The comparison helps evaluate the effectiveness of traditional CNN training versus transfer learning for animal species classification.

### 3 Dataset Description

The dataset used in this project is the **\*\*Animals-10 dataset\*\***, obtained from Kaggle. This dataset contains images of 10 different animal species and is commonly used for image classification tasks. It provides a diverse collection of animal images with different backgrounds, lighting conditions, and image qualities, making it suitable for evaluating deep learning models.

The dataset includes the following animal classes:

Dog  
Horse  
Elephant  
Butterfly  
Chicken  
Cat  
Cow  
Sheep  
Spider  
Squirrel

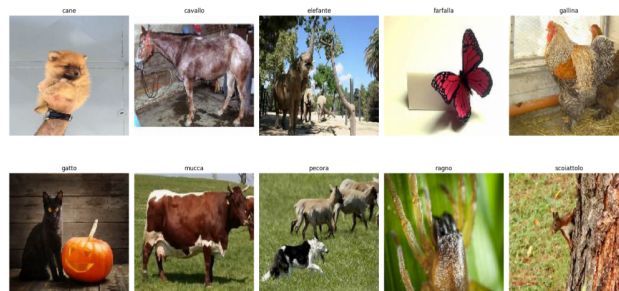


Figure 1: Representative Images from the Animals-10 Dataset.

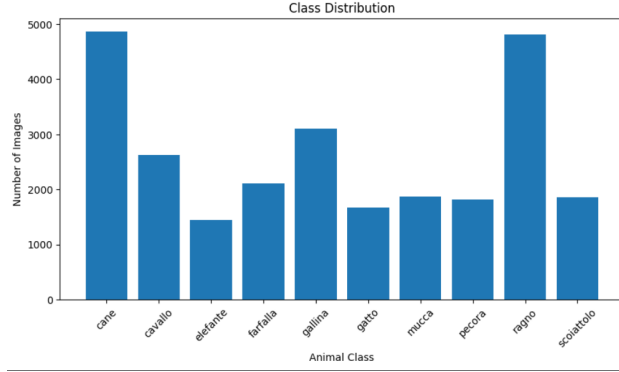


Figure 2: Distribution of Images Across Animal Classes

The total number of images in the dataset is **26,179**. Each class contains a different number of images, which creates slight class imbalance in the dataset. Despite this, the dataset offers sufficient variety for training and evaluating image classification models.

Before training, the dataset was divided into training, validation, and testing sets. This split helps ensure that the models are trained properly and evaluated on unseen data. To prepare the dataset for model training, several preprocessing steps were applied:

- \* Image resizing
- \* Pixel normalization
- \* Data augmentation

All images were resized to a fixed input size to maintain consistency across the dataset. Pixel normalization was applied to improve training efficiency. Data augmentation techniques such as rotation, flipping, and zooming were used to increase dataset diversity and reduce overfitting.

Table 1: Placeholder Caption

|   | Animal Class | Number of Images |
|---|--------------|------------------|
| 0 | cane         | 4863             |
| 1 | cavallo      | 2623             |
| 2 | elefante     | 1446             |
| 3 | farfalla     | 2112             |
| 4 | gallina      | 3098             |
| 5 | gatto        | 1668             |
| 6 | mucca        | 1866             |
| 7 | pecora       | 1820             |
| 8 | ragno        | 4821             |
| 9 | scoiattolo   | 1862             |

This preprocessing step improved the overall quality of input data and helped both models learn more robust features for animal classification.

## 4 Methodology

The methodology followed in this project consists of multiple stages, starting from data preparation to final model evaluation. The main goal was to build an animal species classification system using deep learning and compare the performance of two different models.

The complete workflow of the project includes the following steps:

1. Dataset Collection
2. Data Preprocessing
3. Data Augmentation
4. Model Development
5. Model Training
6. Performance Evaluation
7. Model Comparison

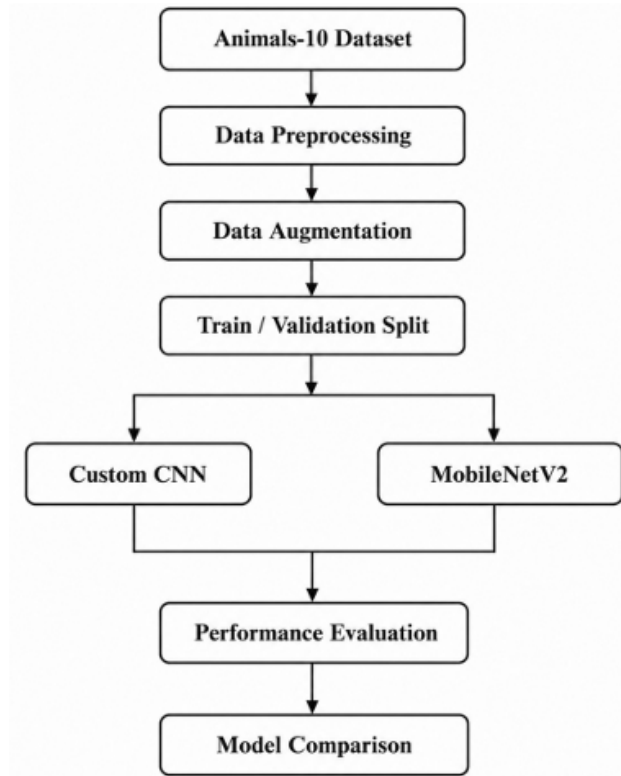


Figure 3: Overall Workflow of the Proposed Animal Species Recognition Framework

The proposed framework was designed to develop and evaluate an automated animal species recognition system based on the deep learning technique. The methodology was divided into five stages: Dataset preparation, Data pre-processing, Development of the model, Training the model and Evaluation of the model's performance. The effectiveness of a custom Convolutional Neural Network (CNN) and the MobileNetV2 transfer learning model was compared through a comparative approach using the same experimental conditions. This

approach would allow for the differences in the performance to be attributed mainly to the model architectures, not differences in data preparation or evaluation procedures.

The Animals-10 data was first uploaded to the Google Colab environment from Google Drive [10]. The data was arranged in folders for each of the 10 animal classes for efficiency when loading it using TF’s Image Data Generator. All images were resized before the training, based on the architectural needs of each model. To simplify the computation, the input images to the custom CNN were  $128 \times 128$  pixels, while the MobileNetV2 used  $224 \times 224$  pixel images as its pre-trained ImageNet model demands this resolution. To make the model

converge faster and to make the optimisation more stable, the pixel values were normalised to a range of 0-1. In order to improve model generalisation, a number of data augmentation techniques were used, but only on the training data. To enhance the diversity of training samples, random rotation, horizontal flipping, zooming, and width and height shifting were employed, but did not change the labels of the classes. This strategy helped minimise the chance of overfitting by introducing more variation in the images to both models, while also maintaining an unbiased validation set to assess each model’s performance.

The first classification model was a custom CNN composed of three convolutional layers, where the ReLU activation function was used after each convolutional layer, and maxpooling operations were performed in order to obtain the features progressively in the spatial dimension. The feature maps obtained from the extraction were flattened and fed into a fully connected dense layer with dropout regularisation, which was then followed by the Softmax output layer to obtain the class probabilities for the 10 different animal classes. This architecture was devised to provide a baseline deep learning model that was trained using the available dataset.

The second model used MobileNetV2 with ImageNet pre-trained weights. The convolutional base was first frozen to retain the learned visual representation, and a new classification head consisting of global average pooling, a dense layer, dropout and a Softmax output layer was added to train this layer for the new task. This transfer learning approach was used to have an efficient adaptation to the Animals-10 dataset and to significantly reduce the time of training and the capability of feature extraction. The Adam optimiser and categorical cross-entropy loss function were used for both models with a batch size of 32. The model performance was assessed using validation accuracy, validation loss, precision, recall, F1 score, classification reports and confusion matrices. These complementary metrics gave a holistic evaluation of the classification efficacy and helped with an objective comparison of the custom CNN and MobileNetV2 transfer learning model.

## 5 Experimental Setup

### 5. Experimental Setup

The experiments for this project were performed using Kaggle Notebook, which provided a cloud-based environment with GPU support for training deep learning models. Python was used as the programming language, and the implementation was carried out using TensorFlow and Keras.

The main libraries used in this project include:

- \* TensorFlow
- \* Keras
- \* NumPy
- \* Matplotlib
- \* Scikit-learn

The dataset was divided into training and validation sets using an 80:20 split. Around 80

To improve model generalization and reduce overfitting, data augmentation techniques were applied to the training images. These techniques included:

- \* Rotation \* Zoom \* Horizontal flipping

The Custom CNN model was trained using resized images with fixed dimensions. The MobileNetV2 model used  $224 \times 224$  pixel images to match the input requirements of the pretrained architecture.

Both models were trained under similar conditions to ensure fair comparison. The Adam optimizer was used for optimization, and categorical cross-entropy was selected as the loss function since this is a multi-class classification problem.

The following configuration was used during training:

\* Development Environment: Kaggle Notebook \* Hardware: Tesla P100 GPU \* Programming Language: Python \* Framework: TensorFlow / Keras \* Dataset: Animals-10 \* Training-Validation Split: 80:20 \* Batch Size: 32 \* Optimizer: Adam \* Loss Function: Categorical Cross-Entropy

This setup provided a consistent environment for comparing the performance of the Custom CNN model and MobileNetV2.

## 6 Results

The performance of both models was evaluated using validation accuracy, validation loss, precision, recall, F1-score, and confusion matrix analysis. Both models were trained on the same dataset and evaluated under identical conditions to ensure a fair comparison.

Custom CNN Performance :

The Custom CNN model achieved a validation accuracy of \*\*67.64

However, the overall performance was moderate, especially for animal classes with similar visual features. Some classes such as dogs, cats, and sheep showed more classification errors compared to visually distinct classes like elephants and butterflies.

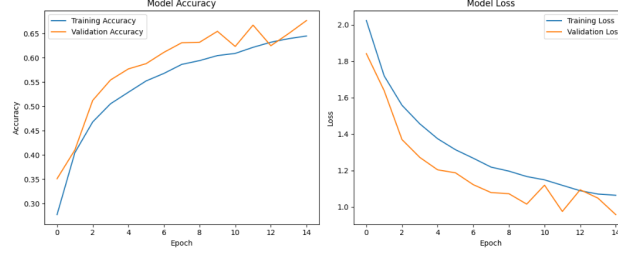


Figure 4: Training and Validation Accuracy and Loss Curves for the Custom CNN

The confusion matrix and classification report indicated that the Custom CNN struggled to differentiate between certain similar-looking animal categories.

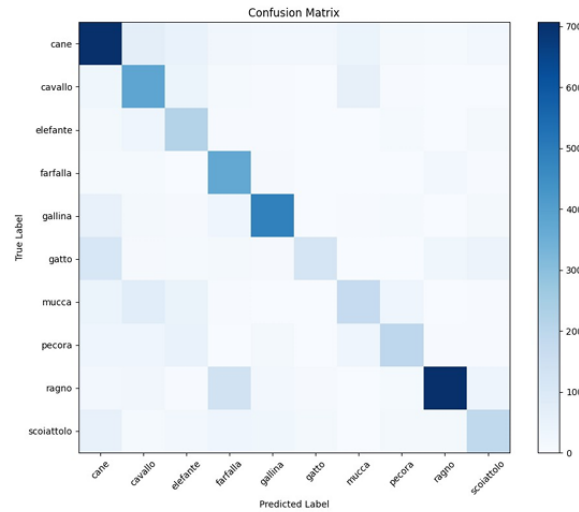


Figure 5: Confusion Matrix of the Custom CNN

MobileNetV2 Performance:

The MobileNetV2 model achieved significantly better performance compared to the Custom CNN. It reached a validation accuracy of \*\*95.39

The model demonstrated excellent classification performance across nearly all animal classes. The confusion matrix showed very few misclassifications, indicating strong feature extraction capability and better generalization.

This improvement highlights the strength of transfer learning in solving image classification problems.

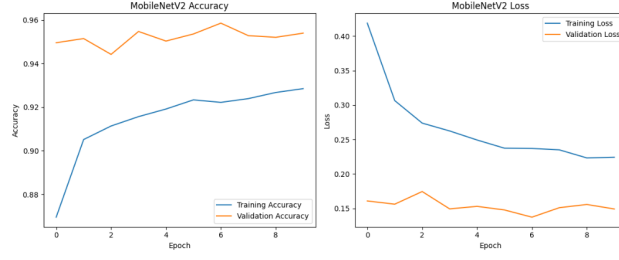


Figure 6: Training and Validation Accuracy and Loss Curves for MobileNetV2

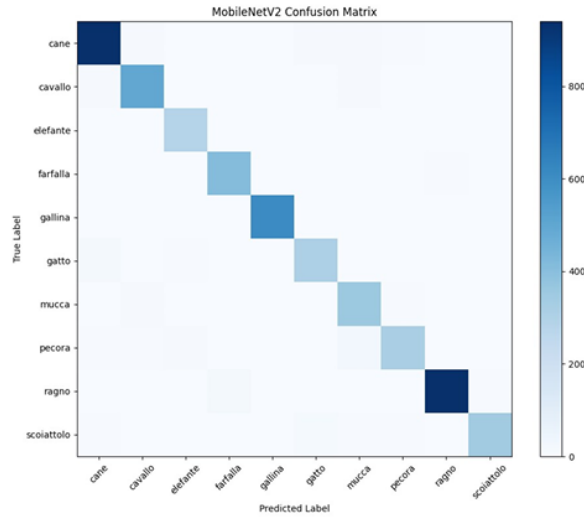


Figure 7: Confusion Matrix of MobileNetV2

Model Comparison:

A comparison of both models clearly shows that MobileNetV2 outperformed the Custom CNN in all major evaluation metrics.

The results confirm that MobileNetV2 provides better feature learning and significantly improves classification accuracy for animal species recognition.

Table 2: Placeholder Caption

| Model      | Validation Accuracy | Validation Loss |
|------------|---------------------|-----------------|
| Custom CNN | <b>67.64%</b>       | <b>0.9572</b>   |
| MobileNetV | <b>95.39%</b>       | <b>0.1491</b>   |

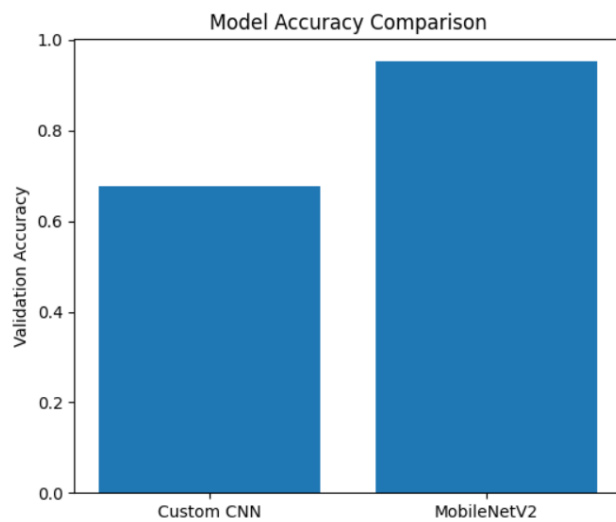


Figure 8: Comparison of Validation Accuracy Between the Custom CNN and MobileNetV2

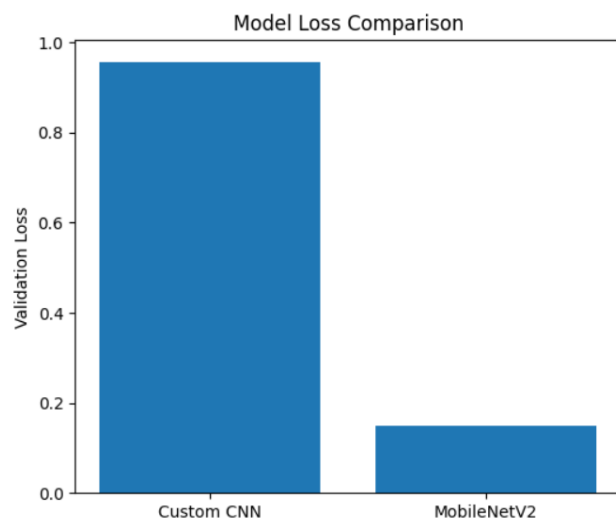


Figure 9: Comparison of Validation Loss Between the Custom CNN and MobileNetV2

## 7 Discussion

The results obtained in this project clearly show that model architecture plays a major role in image classification performance. Although the Custom CNN model was able to learn useful features from the dataset and achieved reasonable performance, its classification accuracy was significantly lower compared to MobileNetV2.

The Custom CNN achieved a validation accuracy of 67.64 which indicates moderate performance for multi-class animal classification. The main limitation of this model was its difficulty in distinguishing animal species with similar visual features. Classes such as dogs, cats, and sheep showed more classification errors due to similarities in shape, texture, and appearance.

In contrast, MobileNetV2 achieved excellent performance with a validation accuracy of 95.39. This large improvement demonstrates the effectiveness of transfer learning in image classification tasks. Since MobileNetV2 was pretrained on ImageNet, it already had strong feature extraction capabilities before being fine-tuned on the Animals-10 dataset.

The superior performance of MobileNetV2 can be attributed to its ability to capture both low-level and high-level visual patterns more efficiently than the Custom CNN model. It also showed better generalization on unseen validation data, resulting in lower loss and fewer misclassifications.

This project highlights that transfer learning is a powerful approach for solving image classification problems, especially when working with limited computational resources or when faster training is required.

The developed system can be useful in practical applications such as wildlife monitoring, biodiversity analysis, ecological research, and automated surveillance systems. Although the results were strong, there are still some limitations in this study. Only one dataset and one transfer learning model were used. Future work could include testing additional architectures such as ResNet, EfficientNet, or DenseNet to further improve performance.

## 8 Conclusion

This project successfully developed and evaluated an animal species recognition system using deep learning techniques. Two different models were implemented and compared: a Custom Convolutional Neural Network (CNN) and MobileNetV2 using transfer learning.

Both models were trained on the Animals-10 dataset and evaluated using multiple performance metrics such as accuracy, loss, precision, recall, F1-score, and confusion matrix analysis.

The experimental results showed that the Custom CNN achieved a validation accuracy

of 67.64 while MobileNetV2 achieved a much higher validation accuracy of 95.39. The comparison clearly demonstrates that MobileNetV2 performed significantly better in terms of classification accuracy, feature extraction, and generalization.

The findings of this project confirm that transfer learning is highly effective for image classification tasks, especially when working with complex datasets containing multiple classes. MobileNetV2 proved to be a reliable and efficient solution for animal species recognition.

In future work, this project can be improved by using larger datasets, exploring advanced transfer learning models, and optimizing hyperparameters to further enhance classification performance.

## 9 References

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